2-Amino-3-(methylamino)-propanoic acid (BMAA) in cycad flour: an unlikely cause of amyotrophic lateral sclerosis and parkinsonism-dementia of Guam.

We conducted an investigation of the levels of the neurotoxin 2-amino-3-(methylamino)-propanoic acid (BMAA) in cycad flour. Analysis of 30 flour samples processed from the endosperm of Cycas circinalis seeds collected on Guam indicated that more than 87% of the total BMAA content was removed during processing. Furthermore, in 1/2 the samples almost all (greater than 99%) of the total BMAA was removed. We found no significant regional differences in the BMAA content of flour prepared from cycad seeds collected from several villages on Guam. Testing of different samples prepared by the same Chamorro woman over 2 years suggests that the washing procedure probably varies in thoroughness from preparation to preparation but is routinely efficient in removing at least 85% of the total BMAA from all batches. Analysis of a flour sample that had undergone only 24 hours of soaking indicated that this single wash removed 90% of the total BMAA. We conclude that processed cycad flour as prepared by the Chamorros of Guam and Rota contains extremely low levels of BMAA, which are in the order of only 0.005% by weight (mean values for all samples). Thus, even when cycad flour is a dietary staple and eaten regularly, it seems unlikely that these low levels could cause the delayed and widespread neurofibrillary degeneration of nerve cells observed in amyotrophic lateral sclerosis and the parkinsonism-dementia complex of Guam (ALS-PD).